

PVE — Stage 4: Exit & Return Simulation

Forward TVPI/DPI/RVPI/IRR distributions · Gaussian-factor copula · assumption sweep · repo ~/dev/pve · 2026-07-05

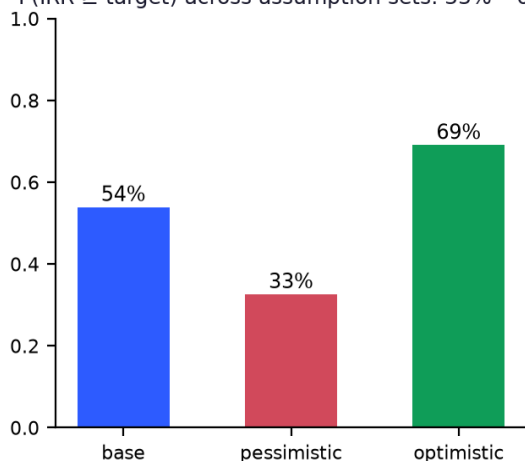
Read this first: what these numbers are, and are not

This stage has **no ground truth**. There is no correct IPO probability to recover, so a single precise-looking number would be confident fiction. Of the 17 calibrated parameter groups, **0 are purely cited, 14 are anchored to published data with assumed interpolation, and 3 are fully ASSUMED** (including the copula loadings — the single most important parameter). Accordingly the headline is a range, led by the multiple: **median TVPI is 1.49x–2.08x across assumption sets** (base 1.79x, 90% CI [0.83, 4.02]; $P(\text{TVPI} \geq 2x) = 38\%$). The rate-based companion — $P(\text{IRR} \geq 15\%) = 33\%–69\%$ — inherits IRR's sensitivity to exit *timing*, which multiple-based targets do not. That swing is the finding.

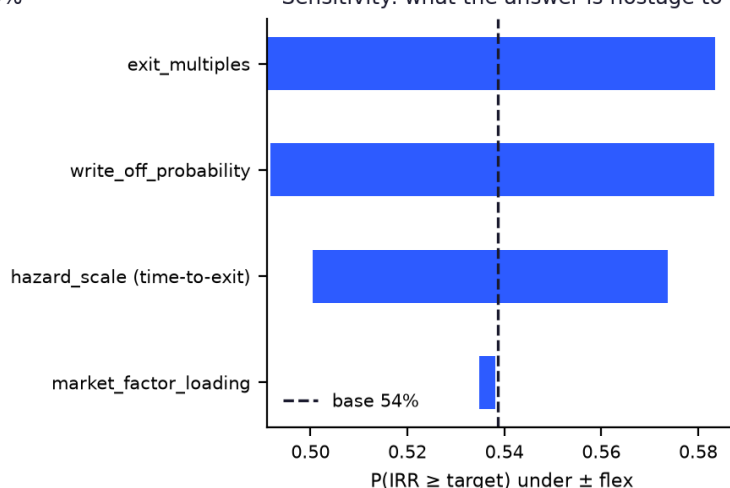
Why TVPI leads: IRR annualizes; an early exit at a high multiple over a fraction of a year produces an enormous rate, so the IRR upper tail (90% bound $\approx 168\%$) is real but annualization-driven and must not be over-read. IRR is reported as median 17.1% with IQR [6.5%, 45.6%]; its mean is deliberately not reported. **Reconciliation note:** component medians do not sum to the TVPI median by construction (medians are percentiles, not additive: $\text{DPI } 1.44 + \text{RVPI } 0.20 \neq \text{TVPI } 1.79$); means reconcile exactly ($1.675 + 0.364 = 2.038$).

The assumption sweep and what the answer is hostage to

P(IRR $\geq$ target) across assumption sets: 33% – 69%



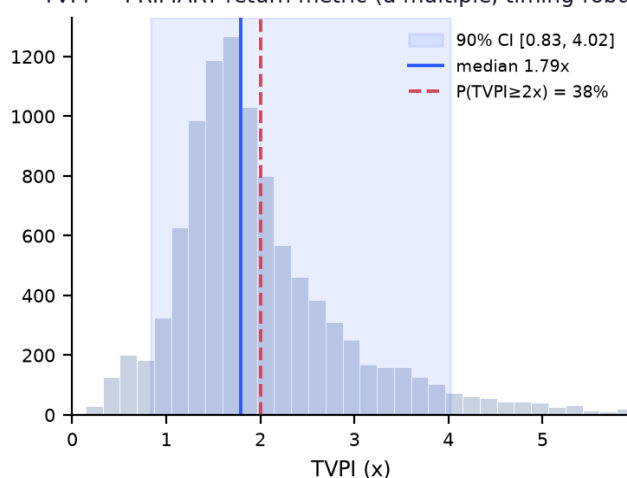
Sensitivity: what the answer is hostage to



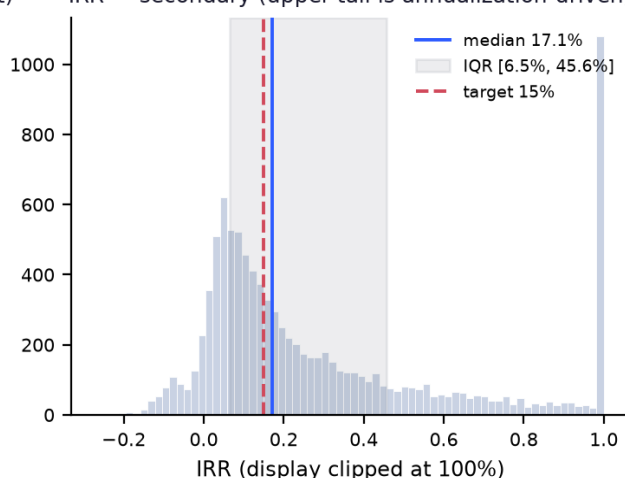
Left: the same simulation under pessimistic / base / optimistic calibrations (same seed — only assumptions differ). Right: one-at-a-time $\pm$ flexes, ranked. Exit multiples and write-off probability dominate; timing and the copula loading move the answer less.

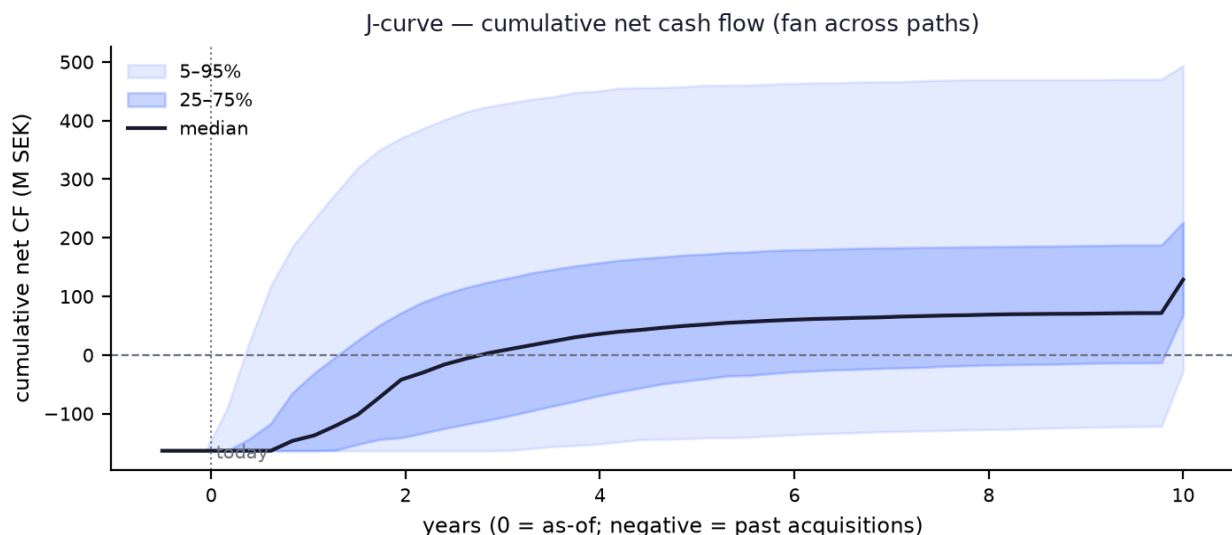
Base-case distributions (with the above caveat attached)

TVPI — PRIMARY return metric (a multiple; timing-robust)



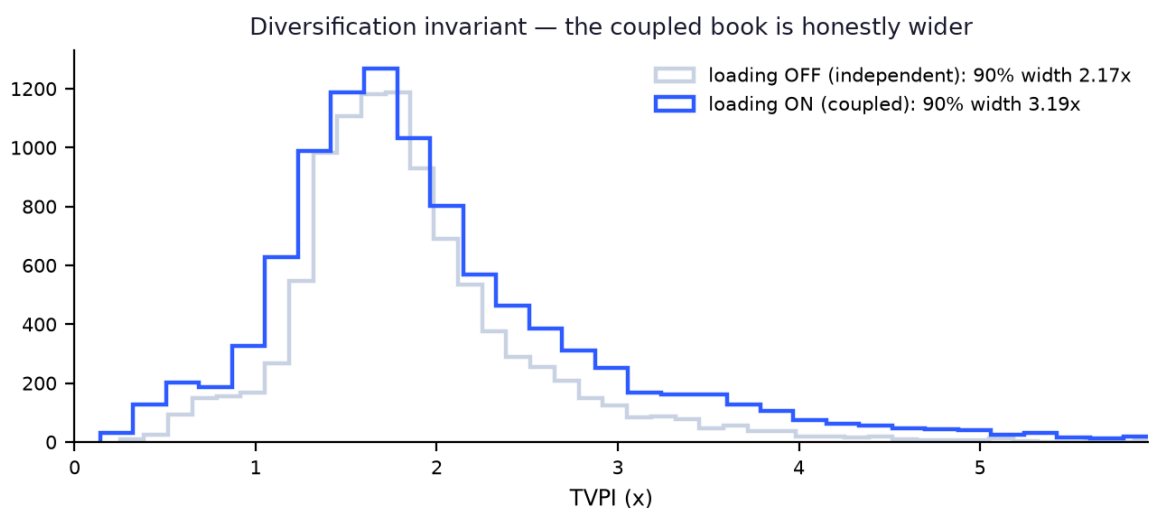
IRR — secondary (upper tail is annualization-driven)





DPI median 1.44, RVPI median 0.20 at the 10y horizon. IRR uses a documented MOIC-implied fallback on 1.0% of paths (no-sign-change streams); no NaN.

The two verifiable parts



1 — Diversification honesty: with the copula loadings zeroed, the book looks better-diversified than it is; the coupled band is wider. **2 — The waterfall mapping is exact:** every exit equity value runs through the Stage-2 payoff curve (verified to 1e-9). In a down-round exit at E=8 on a 10-preference stack, the 1x preference pays the stake 8.0 versus 4.0 pro-rata — naive pro-rata models understate downside protection.

Register audit — every assumption, classified

parameter group	class
exits.probabilities.SEED	ANCHORED
exits.probabilities.SERIES_A	ANCHORED
exits.probabilities.SERIES_B	ANCHORED
exits.probabilities.SERIES_C	ANCHORED
exits.probabilities.GROWTH	ANCHORED
exits.probabilities.PRE_IPO	ANCHORED
exits.weibull.SEED	ANCHORED
exits.weibull.SERIES_A	ANCHORED
exits.weibull.SERIES_B	ANCHORED
exits.weibull.SERIES_C	ANCHORED
exits.weibull.GROWTH	ANCHORED
exits.weibull.PRE_IPO	ANCHORED
exits.multiples.IPO	ANCHORED
exits.multiples.TRADE_SALE	ANCHORED
exits.multiples.SECONDARY	ASSUMED
exits.multiples.DOWN_ROUND	ASSUMED

parameter group	class
exits.copula	ASSUMED

Sources live verbatim in config and the assumptions register (defaults.yaml → exits.assumptions.*). Design calls per spec: Gaussian-factor copula (not regime-switching); multiples-on-last-round (not re-forecasting); invariant-based validation; STAY_PRIVATE marks to the Stage-2 value at horizon. Judgment calls: DOWN_ROUND treated as terminal liquidity at depressed value (continuation is Stage 4.5); SECONDARY is a full exit at a modest multiple (partial sales are Stage 4.5); residual marks are DLOM'd (they are marks), exit proceeds are not (they are cash).